Semester 1 Final Presentation: What We Can Implement

Intelligent Water Network Leak Detection and Localization | IIIT Kottayam | Guide: Dr. Divya Sindhu Lekha

1. Summary

Our slides promise a hybrid hydraulic and data-driven pipeline for detection, topology-informed localization and Top-k ranking, with a Semester 1 scope of 1-2 networks, normal plus single-leak scenarios, varied leak location, size and start time, and an interpretable baseline. The code in the repository delivers part of this: one network (Net3), four leak scenarios, a pressure-threshold detector, a pressure plus flow-divergence localizer, a FastAPI backend and a Next.js dashboard. The professor's feedback that the work is basic matches the gap between the slides and the code. This report lists what can realistically be added before the presentation and in what order, and what should honestly be left to Semester 2.

2. Where we are (checked against the repository after the latest pull)

Area	Status in code	Note
Simulation	Done, one network	Net3 only. 7 scenarios in simulation/scenarios.json: 4 leaks (nodes 3, 15, 115, 121), demand up and down at node 15, pump 335 outage.
Detection	Basic	Fixed 0.5 m pressure-drop threshold against a baseline. It also fires on demand changes and pump outages; the API reports 'cause not classified'.
Localization	Done, simple	Score = peak pressure drop + 50 x positive flow divergence. No graph distance is used anywhere.
Evaluation	Minimal	evaluate.py reports detection count, Top-1, Top-5 and non-leak anomaly flags. No F1, localization error or inference time.
Backend / dashboard	Done	FastAPI and Next.js dashboard with network map and time-series.
Noise, missing data, cross-network	Not done	Planned for Semester 2 on slide 8.

3. Mismatches between the slides and the code (fix before presenting)

- **Slide 7 marks 'Deviation + network distance' as confirmed.** The code uses no network distance. Either add it (section 4, item 3) or mark it as under evaluation.
- **Slide 7 says PostgreSQL storage.** backend/database.py uses SQLite (sql_app.db). Change the slide or switch the database.
- **Slide 7 lists F1, false alarm, localization error and inference time.** Only part of this is computed today.
- **SEMESTER_1_DOCUMENTATION.md claims 100% Top-1.** That was measured on two scenarios. Do not present it as general accuracy; rerun on the larger set and quote those numbers.

4. What we can implement for Semester 1

Ordered by value for the presentation. Items 1-4 are the core and match the Semester 1 scope on slide 8; the rest strengthen it against the 'basic' feedback.

#	Implementation	What it shows	Files
1	Scenario set that varies leak location, size and start time. Sweep a leak at every junction at one size, a size sweep (small to burst) at a few representative nodes, and several start times.	Detection and localization are tested broadly, not on 2-4 cases. Directly the Semester 1 scope on slide 8.	simulation/scenarios.json , generate_dataset.py
2	Second network (Net1 or Net2, from the WNTR examples; Net1-Net6 were in the earlier commit) with its own baseline.	'1-2 representative networks' on slide 8; first evidence of generalisation.	data/networks/, simulation/

#	Implementation	What it shows	Files
3	Add network distance to localization: use shortest-path distance on the pipe graph (networkx is already a backend dependency) to smooth scores over neighbours, and report the error as hops or metres from the true node.	Makes the 'topology-aware' claim true and gives a localization-error metric.	ml_pipeline/localizer.py, evaluate.py
4	Full metrics: detection precision, recall and F1; false-alarm rate on no-leak runs; Top-1, Top-3, Top-5; mean localization error; inference time per scenario. Break results down by leak size and node type (dead-end, hub, high demand).	Quantitative results slide. Shows where the baseline works and fails.	ml_pipeline/evaluate.py
5	Normal vs leak vs other events. Add features that separate them: how widely the pressure drop spreads, whether the loss is concentrated at one node, and unaccounted flow (divergence minus metered demand). Classify an anomaly as leak, demand change or pump/source event. Needs a check that WNTR's demand and leak_demand outputs are usable for this.	Answers the professor's question on telling a leak apart from demand change and equipment change. Needs more event types in the simulator.	simulator.py, new ml_pipeline/features.py, classifier.py
6	More operating-condition scenarios: global demand +/-30%, reservoir head change, tank start level, pump speed, closed pipe. Show the pressure and flow effect of each versus baseline.	Shows how tank, pump and reservoir changes affect the network, as asked.	simulator.py, scenarios.json
7	Adaptive threshold: replace the fixed 0.5 m with a threshold learned from normal variation per network.	Needed for the second network; removes a hard-coded constant.	ml_pipeline/detector.py
8	Dashboard: scenario picker grouped by type, predicted cause, Top-k list with evidence (pressure drop, flow divergence, distance), and a results page.	Live demo that matches the slides.	backend/routers/analysis_api.py, frontend/

5. What to leave for Semester 2

- Sensor noise and missing readings (slide 8 places these in Semester 2). Only a small noise test is worth showing in Semester 1.
- Multiple simultaneous leaks, ML or graph models (autoencoder, One-Class SVM, GNN), and sparse sensor placement using centrality.
- Full cross-network training and testing across 3 or more networks.

6. Suggested order and effort

Step	Work	Estimate
1	Items 1 and 4: bigger scenario set and full metrics on Net3. Gives real numbers first.	2-3 days
2	Item 3: network distance in the localizer and the localization-error metric.	1-2 days
3	Items 6 and 5: operating-condition scenarios, then leak vs demand vs pump classification.	3-4 days
4	Items 2 and 7: second network and adaptive threshold.	2 days
5	Item 8 and slide updates: dashboard changes, fix the slide mismatches, results slides.	2 days

Estimates assume the simulations converge. Large leaks combined with other events may fail to converge in WNTR; test those cases early. Estimates are rough and depend on the presentation date, which I do not know.

7. Suggested presentation structure (add to the existing 9 slides)

- **Scenario design:** table of scenario groups and the 'leak location, size, start time' sweep.
- **Detection results:** detection F1 and false-alarm rate on normal and non-leak runs.
- **Localization results:** Top-1, Top-3, Top-5 and localization error, by leak size and node type, with the network map.
- **Normal vs leak:** one figure showing pressure and flow signatures of a leak, a demand change and a pump outage.
- **Failure analysis:** which cases the baseline misses (small leaks, hub nodes, high demand) and why. This motivates Semester 2.

- **Live demo:** dashboard run on one leak and one non-leak scenario.
- **Semester 2 plan:** noise, missing data, multiple leaks, ML or graph methods, cross-network testing.

All results will come from simulation, not real sensors. State this on the results slides. No new results are reported in this document; the numbers have to be produced by running the evaluation on the extended scenario set.